

Examiner reports

Q1.

(a) was very well answered with around two-thirds of students scoring full marks.

(b) proved to be quite a challenging question with a wide spread of marks. A number of students failed to see the connection to (a) despite the “Hence...” in the question, and so never arrived at an appropriate integrand to integrate. There were a wide variety of methods used to integrate including inspection and various substitutions which had the potential to be successful, and some very imaginative approaches were seen. Some students used a substitution of $x = 2\theta$ initially but failed to correctly replace $d\theta$, meaning that, despite correct later work, they were unable to arrive at a valid, correct solution.

Q3.

In Part (a), most students gained the method mark by using the identity $\sec^2 x = 1 + \tan^2 x$ and many went on to earn the accuracy marks. A significant number of students lost the final accuracy mark through poor notation such as writing $\tan x^2$ instead of $\tan^2 x$. It was

surprising to see some students working at this level to write $\frac{\sec^2 x}{\sec^2 x - 1}$ as $\frac{\sec^2 x}{\sec^2 x} - \frac{\sec^2 x}{1}$.

Part (b) proved difficult for most students. Many students assumed that they could factorise the quadratic equation and they were satisfied with factors that were clearly incorrect, such as $(\operatorname{cosec} x - 3)(\operatorname{cosec} x + 1)$. Some students misquoted the quadratic formula; some others muddled factors and roots. Most students showed a correct conversion from $\operatorname{cosec} x$ to $\sin x$ after obtaining their roots but those who attempted to convert their equation at the start rarely proceeded to a correct quadratic equation in $\sin x$. Some students discarded the negative root of their quadratic equation.

In part (c), the students who answered part (b) correctly usually gained both marks for this part, although some lost the accuracy mark for not writing their answers to the nearest degree. Where both marks were not earned, the majority of the students gained the method mark for equating $2\theta = 60^\circ$ to one of their solutions in part (b).

Q4.

Expansion of the brackets was tackled in most cases, but with many sign and numerical slips, particularly failing to square the “3”. Most candidates knew they might need to use the identity $\sin^2 \theta + \cos^2 \theta = 1$, but did not always use it in a constructive fashion. Giving θ a numerical value to find the relevant integer earned no credit in this trigonometrical proof question.

Q5.

There was a full range of marks seen in response to this question. Some candidates were able to complete the whole question successfully, including finding all four solutions to the particularly demanding part (a)(ii). Other candidates just scored 1 or 2 marks, from part (a)(i) and/or a correct identity in part (b)(i). Most candidates knew what to do for part (a)(i)

and got the value of R correct, although a few got $\sqrt{80}$. Many, however, had $\tan \alpha = \frac{6}{8}$.

Most also knew that they were to use the expression from part (a)(i) in part (a)(ii), although a few attempted to manipulate the given expression with double angle formulae and usually abandoned the attempt. Of those using the expected approach, most scored 2 marks, some getting a third solution as well but not many finding all four. An error which was sometimes seen was to replace $2x + 53.1$ with $\theta + 106.2$.

Part (b) was generally more successful, with most candidates scoring a mark for at least one of the identities for $\sin 2x$ or $\cos 2x$ correct and many going on to use these convincingly in obtaining the requested result. Others made errors in signs and coefficients and fudged the result rather than looking for their errors. Some candidates, however, did seek an error and attempted to overwrite their work rather than starting again, often making it difficult to read what they had actually intended as their answer.

Most candidates who attempted part (b)(ii) scored at least 2 marks — the starting equation and the solution $x = 45$ — but many also obtained all four solutions. Those candidates who ignored the hence and again expanded the left-hand side were largely unable to complete the question successfully, although some did succeed via this route.

Q6.

It was notable in this question that candidates tended to be successful in one part or the other but not both, although some good fully correct solutions were seen.

In part (a) many clear derivations of the given result were seen. These often involved some unnecessary steps, such as returning to double angles to eliminate $\cos^2 x$. Those who didn't complete this part successfully usually had an error in the double angle formulae, often omitting a 2. Many candidates wrote down the given answer even though this clearly did not follow from the previous line.

For part (b) most candidates knew they were expected to use the result from part (a) to do this integral; although some just ignored it, with $\sin^4 x$ often appearing in such solutions. Few correct alternative methods to the one expected were seen. The common errors in the expected method were in sign and coefficient errors in the integrals of $\sin x$ and $\sin 3x$.

Q7.

Most candidates just wrote down the required expression for $\cos 2x$, although some worked it out from $\cos^2 x - \sin^2 x$ or $\cos(A + B)$. Omission of the 2 was a relatively rare error. Most candidates were able to derive the result in part (b)(i) although some were penalised for being too brief. Most candidates too were able to solve the resulting quadratic equation, with some making it a little more difficult than intended by reading the 1 on the right hand side as 0. Most candidates found two results for angle x in the required range. Part (c) was less successful: although most candidates understood that they needed the integral in terms of $\cos 2x$, there were many algebraic slips, and sign and coefficient errors in the integral of $\cos 2x$ itself.

Q8.

This question proved to be more straightforward than anticipated, with many candidates gaining full marks from a well-presented answer. Most candidates could do part (a), although a few did not simplify their answer, although they were not required to. However $\tan 2x$ expressed as $\tan(2 + x)$ was seen a few times. In part (b) most candidates took the

anticipated route of substituting for $\tan 2x$, and then convincingly dealt with the resulting algebraic fraction. Some then made a sign error in their simplification. Some manipulated the expression for $\tan 2x$ into an expression for $\tan^2 x$ and then substituted that into the expanded form of $(1 - \tan x)^2$. Some did this correctly, but others got bogged down due to algebraic errors and often abandoned their attempt.

Q9.

Part (a) was very well answered by the majority of candidates although errors of $\tan^2 x$ and $\ln(\sec x)$ were seen.

In part (b), the first step, the use of the quotient rule, was reasonably well done although the quotient rule was not always quoted correctly. Errors occurred where candidates had wrong signs for derivatives of $\sin x$ and $\cos x$ or where the functions in the numerator were reversed. Many candidates went on and used appropriate trigonometric identities to obtain the given answer.

Part (c) was not very well answered by most of the candidates. Many candidates obtained the correct expansion of $(\tan x + \cot x)^2$ but were then unable to handle the term $2 \tan x \cot x$. There were numerous flawed attempts using expressions such as $(\tan^2 x + 1) + (\cot^2 x - 1)$ equals the answer given.

In part (d), many candidates obtained the first method mark but went no further. Those who correctly integrated the expression often lost the final accuracy mark by working in degrees, with 57° being common.

Q10.

Most candidates were able to gain some marks in part (a). In (i) the common wrong answer was a stretch in the x -direction with scale factor $1/2$. In (ii) most gave a correct reflection and in (iii) many gave the correct vector or its equivalent but still some candidates are not using the correct word 'translation'. As previously stated, 'tr' was not accepted as a replacement for 'translation'.

Part (b) was not answered well. The usual common error, illustrated by ' $\theta - 30 = 44.4$, $\theta = 74.4$ or $\theta = 180 - 74.4$ ' was seen too frequently. It was not uncommon to see candidates giving answers in all four quadrants. A more surprising error from the better candidates is illustrated by ' $\theta - 30 = 44.4$, 135.6 so $\theta = 14.4$, 105.6 '

Part (c) was answered well but the usual errors $(\cos x + \sin x)^2 = \cos^2 x + \sin^2 x$ and $(\cos x + \sin x)^2 = \cos^2 x + \sin^2 x + \cos x \sin x$ and the 'trig identity ' $\cos x + \sin x = 1$ ' were all seen.

Q11.

This question in general was not done well. A lot of candidates indicated a poor knowledge of trigonometric identities and a lack of experience in manipulating them. The Specification states that the double angle formulae should be learnt.

Part (a) Although most candidates could recollect or work out the identity for $\sin 2x$ this was not the case for $\cos 2x$ where many variations on incorrect answers were seen; some

were correct but not in the requested form.

Part (b) Candidates were given credit for attempting to use their versions of $\sin 2x$ and

$\cos 2x$ together with $\tan x = \frac{\sin x}{\cos x}$. Those who started on the left hand side often omitted a common denominator of $\cos x$, some just putting it back into their work to get to the given answer. Those who started on the right hand side were generally more successful, but most candidates worked from both sides together, often making errors in their manipulations, such as omitting brackets and thus not multiplying out correctly. Many managed to make the two sides become equal after some invalid manipulation.

Part (c) Most candidates made the solution of this equation far more involved than was intended or is necessary. Few realised they could use the answer from part (b) and make use of, if $pq = 0$ then $p = 0$ or $q = 0$ and thus write down the solutions, with little more work. Some candidates did write down some correct solutions with no working seen, which was accepted. Many manipulated either form of the equation, into sines and cosines often making errors and abandoning the attempt. Many derived the cubic equation $\sin x - 2\sin^3 x = 0$ or something close to it, but still did not realise that $\sin x = 0$ was a solution. Of those who did get to correct solutions, the solution 225° , was often missing.



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